



SD-1.60 – Static & Dynamic Balancing Apparatus

A bench mounted enclosure contains all the electronics and safety guard for this apparatus.

On top is a shaft running in bearings. The shaft can carry up to four balance masses (non-coplanar). Each mass can be easily adjusted for imbalance by its radius, linear and angular position. Linear and angular scales ensure accurate positioning of each mass. A set of 12 balance masses are provided of varying mass, and each one has a thin alignment slot produced.

PRODUCT CODE: SD-1.60

Categories: Educational, SD-1.60-Static & Dynamic Balancing Apparatus, SV - Theory Of Machines, SV Series - Statics & Vibrations

Tags: Angular Scales , Balance Masses , Linear Scales , Non-Coplanar , Precision Weights , Static & Dynamic Balancing Apparatus , SV Series

DESCRIPTION

A bench mounted enclosure contains all the electronics and safety guard for this apparatus.

On top is a shaft running in bearings. The shaft can carry up to four balance masses (non-coplanar). Each mass can be easily adjusted for imbalance by its radius, linear and angular position. Linear and angular scales ensure accurate positioning of each mass. A set of 12 balance masses are provided of varying mass, and each one has a thin alignment slot produced.

During dynamic balancing the shaft is run from a speed controlled motor through a pulley and belt arrangement. The speed of the motor is accurately controlled from the front panel. When static balancing is required the shaft is disengaged from the motor by simply sliding off the belt.

To statically balance the shaft and each individual balance mass, the balance moment is measured using precision weights, bearing balls, pulley and cords.

TECHNICAL SPECIFICATIONS

Speed controlled motor shaft: 0-1400rpm

Balance mass radius: 30-70mm

Balance masses: 50, 75, 100g

Linear adjustment: Scale in millimetres

Angular adjustment: 360°, 1° resolution

FEATURES & BENEFITS

- Self-contained bench top apparatus
- All rotating parts fully guarded
- Three mass sets of different mass
- Masses can be adjusted for radius
- Many experimental adjustments can be made
- Speed controlled and limited
- Power 'cut-out' with guard removed

Experimental Content

- Static balancing of non-coplanar masses
- Dynamic balancing of non-coplanar masses
- Comparison between theoretical and actual results
- Use of vector diagrams, vector resolution, resultant forces, moment polygons and turning moments

Related Laws

- Static Balancing
- Dynamic Balancing
- Non-Coplanar Masses
- Vector Diagrams
- Vector Resolution
- Resultant Forces
- Moment Polygons
- Moments
- Imbalance
- Vibration

Ordering Specification

- 1 x SD-1.60
- 1 x Protective dome
- 1 x Static balancing attachment
- 4 x 50g Balance mass
- 4 x 75g Balance mass



- 4 x 100g Balance mass
- 100 x Ball weights
- 2 x Load hanger
- 1 x Combination spanner
- 1 x Power supply
- Locking handle
- Instruction manual
- Packing list
- Test sheet

Requirements

Electrical supply: 110/120V, 60Hz or 220/240V, 50Hz

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 0.24m³

Gross weight: 31kg

Dimensions

Length: 0.500m

Width: 0.500m

Height: 0.460m

Order Codes

SD-1.60-A Static & Dynamic Balancing Apparatus

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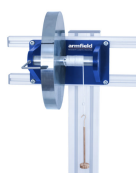
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